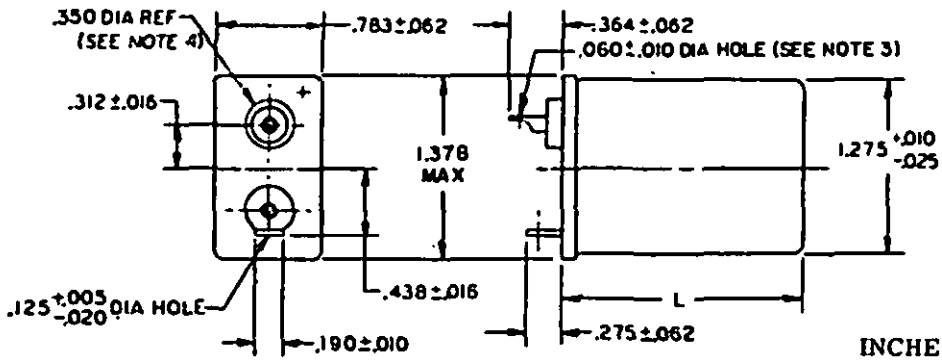
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, ETCHED FOIL), 85°C (VOLTAGE DERATED TO 125°C), STYLE CL53

INACTIVE FOR NEW EQUIPMENT DESIGN. NO SUPERSEDING DOCUMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

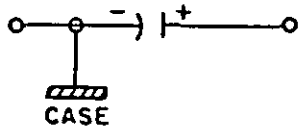
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.



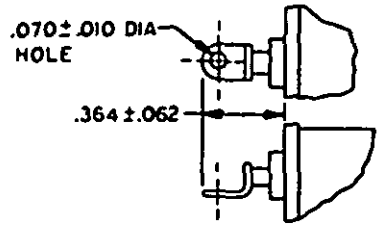
TYPE D CASE

Dimensions	
	L ±.062 (1.57)
D1	1.375 (34.93)
D2	2.000 (50.80)
D3	2.500 (63.50)

INCHES	MM	INCHES	MM
.005	.13	.190	4.83
.010	.25	.275	6.99
.016	.41	.312	7.92
.020	.51	.350	8.89
.025	.64	.364	9.25
.060	1.52	.438	11.13
.062	1.57	.783	19.89
.070	1.78	1.275	32.39
.125	3.18	1.378	35.00



CIRCUIT DIAGRAM



OPTIONAL ALTERNATE TERMINAL CONSTRUCTION

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Orientation of terminal hole not fixed.
4. Glass to metal (hermetic) seal.

Ⓗ FIGURE 1. Style CL53 capacitors.

Ⓗ denotes changes

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Ⓗ TABLE I. Capacitor characteristics.

Type designation	Voltage			Capacitance	DC leakage (max)			Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C	125°C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	µA	ohms	
CL53BE122UPG	15	10	17.2	1,200	42	335	420	7	D1
CL53BE242UPG	15	10	17.2	2,400	84	670	840	4	D2
CL53BE352UPG	15	10	17.2	3,500	122	975	1,220	2	D3
CL53BH661UPG	30	20	34.5	660	46	370	460	13	D1
CL53BH132UPG	30	20	34.5	1,300	75	600	750	6	D2
CL53BH182UPG	30	20	34.5	1,800	92	735	920	5	D3
CL53BJ351UPG	50	30	57.5	350	36	290	360	11.5	D1
CL53BJ701UPG	50	30	57.5	700	73	585	730	5.7	D2
CL53BJ102UPG	50	30	57.5	1,000	105	840	1,050	4.0	D3
CL53BL201UPG	75	50	86.2	200	35	280	350	20	D1
CL53BL401UPG	75	50	86.2	400	70	560	700	10	D2
CL53BL551UPG	75	50	86.2	550	96	770	960	7.2	D3
CL53BI121UPG	112	75	129.0	120	31	250	310	86	D1
CL53BI241UPG	112	75	129.0	240	63	505	630	16.5	D2
CL53BI321UPG	112	75	129.0	320	84	670	840	12.5	D3
CL53BQ700UPG	150	100	172.0	70	24	190	240	57	D1
CL53BQ141UPG	150	100	172.0	140	49	390	490	29	D2
CL53BQ201UPG	150	100	172.0	200	70	560	700	20	D3

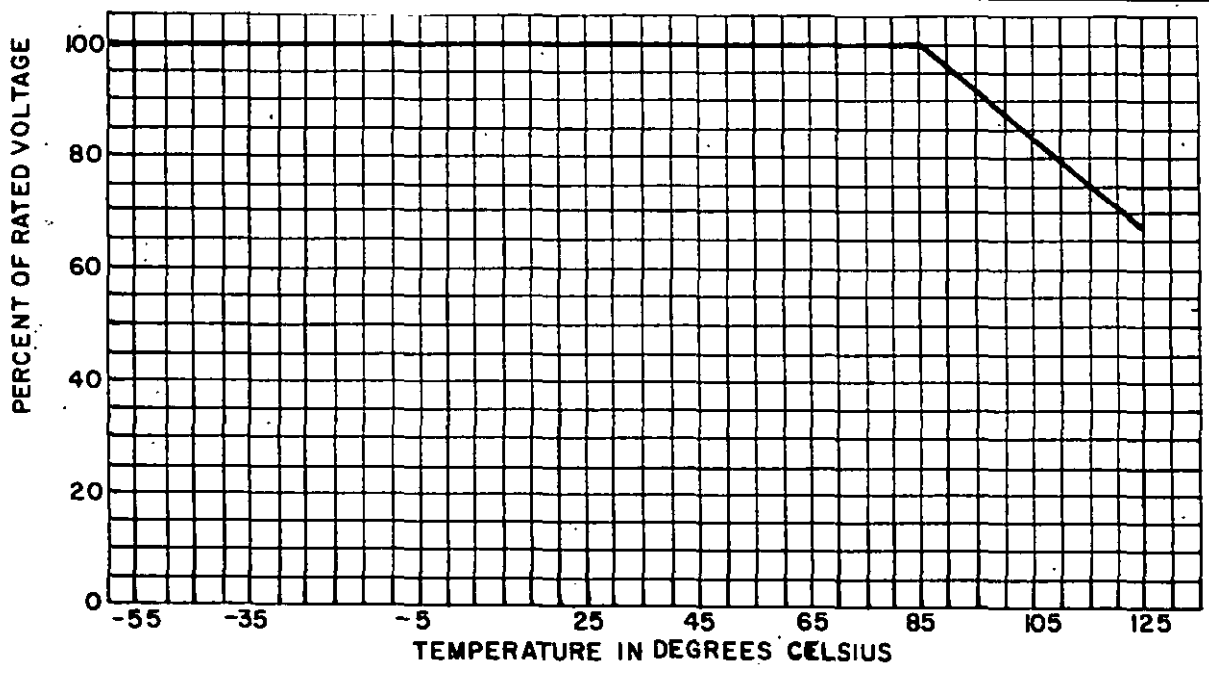


FIGURE 2. Voltage derating with temperature.

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REQUIREMENTS:

Design and construction:
Dimensions and configuration:
Capacitors - See figure 1.
Brackets - See figure 3.
Case type - Rectangular, uninsulated.
Seal type - Hermetic (symbol G).
Terminals - See figure 1.
Rated temperature - -55° C to +85° C, voltage derated to +125° C (see figure 2).
Rated voltage - See table I.
Capacitance (cap) (nom) - See table I.
Capacitance tolerance - Symbol U (+75, -15%).

DC leakage (DCL) (max) (at +25° C):
See table I.

Dissipation factor (DF) (max): 30% (15 v), 20% (30 thru 75 v), or 15% (112 and 150 v).

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - Not applicable.
Terminal to case - Not applicable.

Insulation resistance:
Sleeving to case (min): Not applicable.
Terminal to case - Not applicable.

Temperature cycling:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - 115% of initial requirement.

Solderability: Not applicable.

Terminal strength:
Pull test - In accordance with MIL-C-3965.
Bend test (if req.) - In accordance with MIL-C-3965.
Torque test - Not applicable.
Wire-lead bend test - Not applicable.
Twist test - Not applicable.

Surge voltage: See table I.

Moisture resistance:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±6% of initial measurement.
DF (max) - 115% of initial requirement.

Low temperature (storage):
DCL (max) - See table I (25° C).
ΔCap - Within ±6% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Stability at low and high temperature:
Step 1 (+25° C):
DCL (max) - See table I.
Cap - Within tolerance of table I value.
DF (max) - Shall not exceed initial requirement.
Step 2 (-55° C):
Impedance - See table I.
ΔCap (max) - -65% (≤30 v) or -45% (>30 v) of step 1 value.
Step 3 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85° C):
DCL (max) - See table I.
ΔCap (max) - +75° of step 1 value.
DF (max) - 50%.
Step 5 (+125° C):
DCL (max) - See table I.
ΔCap - +75% of step 1 value.
DF (max) - 50%.
Step 6 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

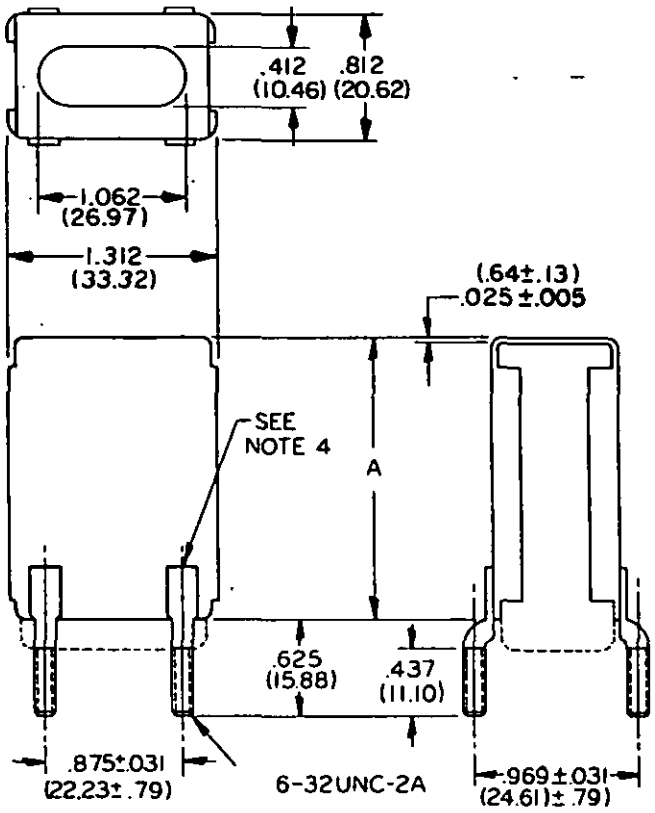
Reverse voltage: Not applicable.

Life at +85° C:
2,000-hr (qualification):
DCL (max) - See table I (25° C).
ΔCap - Within +15, -20% of initial measurement.
DF (max) - 130% of initial requirement.
240-hr (group B):
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - 130% of initial requirement.
1,760-hr (group C):
DCL (max) - See table I (25° C).
ΔCap - Within +15, -20% of initial measurement.
DF (max) - 130% of initial requirement.

Barometric pressure: Rated voltage.

Life at +125° C:
DCL (max) - See table I (25° C).
ΔCap - Within ±25% of initial measurement.
DF (max) - 150% of initial requirement.

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BRACKETS FOR TYPE D CASE

Type designation	Case size	Dim	Dimensions				
			Inches			Millimeters	
			Min	Nom	Max	Min	Max
CL050SD13	D1	A	1.159	1.175	1.191	29.44	30.25
CL050SD23	D2	A	1.784	1.800	1.816	45.31	46.13
CL050SD33	D3	A	2.284	2.300	2.316	58.01	58.82

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch ≡ 25.4 mm.
4. Spade lugs shall be rigid (either double welded, double riveted, single riveted and soldered, or single riveted and welded to the bracket).
5. Unless otherwise specified, tolerance is ±.062 (1.57 mm).

FIGURE 3. CL050 bracket.

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Marking: Capacitors shall be marked as shown in the following example:

Example:

CL53BE122UPG - Type designation.
12345 7015 - Source and date code.
1,200 μF 15 vdc - Capacitance and rated voltage (at 85° C).

Polarity: Plus (+) symbol near positive terminal (see figure 1).

Voltage groups (85° C rated voltage):

I - - - - - - - - - - - - 15 to 75 v, incl.
II - - - - - - - - - - - - 112 and 150 v.

Substitution data. Capacitors covered by this specification are directly substitutable for capacitors of equivalent capacitance, capacitance tolerance, and voltage in the same case size.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Navy - SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Air Force - 85

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